Quantitative Retrieval and Classification from Multi-Sensor Optical Imagery:

Bathymetric Estimation, Multi-Resolution Image Fusion, and Spectral Classification in Coastal and Wetland Environments

Mohammed Abo Oraig
University of Toulon
mohammed-abo-oraig@etud.univ-tln.fr

Abstract

Optical Earth observation imagery is increasingly used to derive quantitative environmental information in settings where in-situ measurement is costly or unavailable, but such retrieval is complicated by heterogeneous spatial resolution, atmospheric and surface-reflection artefacts, and the need to translate spectral signal into physical or categorical variables. This paper presents four case studies conducted as part of a practical training in aerospace image processing, each addressing a distinct aspect of this problem using a different optical sensor. First, water-column depth over a coastal shallow-water area (Porquerolles, France) is estimated from Pléiades multispectral imagery using a logarithmic blue-to-green band-ratio index, with a linear-regression-based correction applied to remove the sun-glint and atmospheric contribution captured in the near-infrared band. Second, a Landsat 8 OLI scene is used to increase the spatial resolution of a multispectral composite by injecting high-resolution panchromatic detail, comparing the Brovey and HSV fusion methods. Third, Sentinel-2 MSI imagery of the Camargue wetland is classified into soil, vegetation, and water classes using region-of-interest spectral signatures compared via Euclidean distance and the Spectral Angle Mapper. Fourth, Sentinel-3 OLCI imagery is used to detect anomalous chlorophyll-a signal associated with floating algae, isolating local anomalies from the slowly varying background reflectance via a large-kernel median filter. Across all four studies, the workflow follows a common structure — band loading and compositing, radiometric/spectral index construction, thresholding or classification, and cartographic visualization — implemented in Python using NumPy, OpenCV, and SciPy. The results illustrate practical competency in deriving quantitative and categorical geospatial information from optical imagery of differing spatial and spectral characteristics, while also surfacing methodological limitations — absence of independent validation, assumption-based depth calibration, and unquantified threshold sensitivity — that are discussed as directions for more rigorous follow-up work.

1 Introduction

Optical remote sensing has become a standard tool for environmental monitoring, offering repeated, spatially continuous observation of land and water surfaces at a range of spatial, spectral, and temporal resolutions. Translating this raw spectral signal into scientifically or operationally useful information, however, is rarely straightforward: sensors differ in spatial resolution and spectral band configuration, the atmosphere and specular reflection at water surfaces introduce a signal component unrelated to the surface of interest, and the boundary between two thematic categories (e.g. water versus land, healthy vegetation versus soil) is often gradual rather than sharp. Addressing these issues — extracting a physical quantity, a higher-resolution product, or a discrete category from imperfect

and heterogeneous imagery — is a recurring methodological challenge across many applications of Earth observation, from coastal management to land-cover monitoring.

This paper reports four case studies that each address one facet of this problem using a different optical sensor and a different target variable:

1. **Bathymetric retrieval.** Can water depth in a shallow coastal zone be estimated from a multispectral image alone, given that the raw reflectance signal is confounded by sun-glint and atmospheric scattering? Using Pléiades imagery of Porquerolles, we test a logarithmic blue-to-green band-ratio index, combined with a regression-based correction that uses the near-infrared band — which does not penetrate water — as a proxy for the surface-reflection component to be removed from the visible bands.

2. **Multi-resolution image fusion.** Can the spatial detail of a high-resolution panchromatic band be transferred to a lower-resolution multispectral composite while preserving radiometric fidelity to the original colours? Using a real Landsat 8 OLI scene, we compare two classical pan-sharpening approaches — the Brovey transform and HSV-based substitution — and assess their respective trade-offs.

3. **Spectral classification of a heterogeneous landscape.** Can distinct land-cover classes (soil, vegetation, water) in a wetland environment be separated using per-pixel spectral signatures alone? Using Sentinel-2 MSI imagery of the Camargue, we compare a Euclidean-distance classifier against the Spectral Angle Mapper (SAM), which is invariant to signal magnitude and therefore, in principle, more robust to illumination differences.

4. **Anomaly detection against a spatially varying background.** Can a highly localized surface phenomenon — floating algae — be isolated from natural, slowly varying spatial variation in ocean colour? Using Sentinel-3 OLCI imagery, we compute the Maximum Chlorophyll Index (MCI) and subtract a smoothed background estimate (obtained via a large-kernel median filter restricted to water pixels) to obtain a differential index (dMCI) that highlights local anomalies.

Across the four case studies, a common processing structure is followed: multi-band image loading and compositing, radiometric normalization for display and, where relevant, for quantitative computation, construction of a spectral index or reference signature, and a final classification, thresholding, or fusion step whose output is interpreted and visualized cartographically. All processing was implemented in Python (NumPy, OpenCV, SciPy).

Optical band-ratio methods for estimating water depth from passive multispectral imagery were first proposed by Lyzenga (1978) and later refined into a two-parameter logarithmic ratio form by Stumpf et al. (2003), which is the form most directly comparable to the bathymetric index used in Section 3.1. Pan-sharpening methods such as the Brovey transform and intensity-hue-saturation (IHS/HSV) substitution, used in Section 3.2, are reviewed and formally compared by Pohl and van Genderen (1998). The Spectral Angle Mapper used in Section 3.3 was introduced by Kruse et al. (1993). The Maximum Chlorophyll Index used in Section 3.4 was introduced by Gower et al. (2005) for the detection of intense plankton blooms from the MERIS 709 nm band.

This paper focuses on these four studies because each involves a quantitative or classification-based transformation of spectral data under resolution or signal-quality constraints — a problem structurally similar to that of extracting reliable metrics from heterogeneous, data-scarce remote-sensing sources more broadly. The broader practical training of which this work is part also included four further applications not detailed here for reasons of space: bi-temporal flood-extent mapping in the Var department following the November 2019 rainfall event, using near-infrared thresholding

and image differencing; burn-scar mapping following the 2017 Var wildfire, using NDVI thresholding and bi-temporal differencing; object counting, applied to detect seasonal vessel-traffic variation around Port-Cros via connected-component labelling of a binarized image; and the acquisition and classification of field spectra (vegetation, soil, plastic, rock) using a handheld spectrometer and ENVI, applied to a hyperspectral scene of a beach environment. These additional exercises reinforced the same underlying skills — spectral index construction, thresholding, bi-temporal comparison, and classification — applied to different sensors, targets, and time scales.

2 Data

The four case studies draw on four different optical sensors, spanning spatial resolutions from 2 m to 300 m and covering distinct environments in southern France: a shallow coastal shelf (Porquerolles), an unspecified terrestrial scene acquired by Landsat 8, a coastal wetland (Camargue), and open coastal waters (Mediterranean, exact scene extent not specified in the original material). A summary of all datasets is given in Table 1; sensor-specific details follow in Sections 2.1–2.4.

2.1 Bathymetric retrieval: Pléiades imagery of Porquerolles

Water-depth retrieval was performed on a Pléiades multispectral image covering the coastal shelf around the island of Porquerolles (Var, France). The image comprises four spectral bands (blue, green, red, near-infrared) at 2 m spatial resolution.

2.2 Multi-resolution fusion: Landsat 8 OLI scene

Image fusion was demonstrated on a real Landsat 8 OLI scene, using the panchromatic band (native 15 m resolution) and a three-band (blue, green, red) multispectral composite (native 30 m resolution), giving a resolution ratio of 2 between the two products. The multispectral and panchromatic products have 512×512 and 1024×1024 pixels respectively, with resolutions reported as 30 m and 15 m.

2.3 Land-cover classification: Sentinel-2 MSI imagery of the Camargue

Land-cover classification was performed on a Sentinel-2 MSI image of the Camargue wetland, sourced via the Copernicus Browser. Four spectral bands at 10 m spatial resolution were used: B02 (blue, 490 nm), B03 (green, 560 nm), B04 (red, 665 nm), and B08 (near-infrared, 842 nm). The product was delivered at processing level L2A (atmospherically corrected, surface-reflectance) and downloaded as 8-bit TIFF.

2.4 Algae anomaly detection: Sentinel-3 OLCI imagery

Floating-algae detection used a Sentinel-3 OLCI image acquired on 8 August 2018, processed at level L2 (geolocated, atmospherically corrected). OLCI provides 21 spectral bands at 300 m spatial resolution; five bands were used, corresponding to the spectral region critical for chlorophyll-a fluorescence detection: B03 (442 nm, blue), B06 (560 nm, green), B10 (681.25 nm), B11 (708.75 nm), and B12 (753.75 nm, near-infrared).

3 Methods

Across all four case studies, imagery was loaded as multi-band arrays, composited into true-colour (RGB) and false-colour (NIR–G–B) renderings for visual inspection, and processed using NumPy,

Table 1: Summary of datasets used across the four case studies.

Case study	Sensor	Bands used	Resolution	Processing level
Bathymetry	Pléiades	4 (B,G,R,NIR)	2 m	Not specified
Fusion	Landsat 8 OLI	PAN + 3 (B,G,R)	15 m / 30 m	Not specified
Classification	Sentinel-2 MSI	4 (B02,B03,B04,B08)	10 m	L2A
Algae detection	Sentinel-3 OLCI	5 (B03,B06,B10–B12)	300 m	L2

OpenCV, and SciPy in Python. Case-study-specific processing is described below.

3.1 Bathymetric retrieval (Pléiades)

Water depth was estimated from the blue and green Pléiades bands using a logarithmic band-ratio index, following the general principle that light penetration depth in water decreases with wavelength, so that the blue-to-green reflectance ratio decreases systematically with depth (Lyzenga, 1978; Stumpf et al., 2003):

$$X = \frac{\ln(n \cdot R_{\text{blue}})}{\ln(n \cdot R_{\text{green}})} \tag{1}$$

where R_{blue} and R_{green} are the blue and green band reflectance values and n is a positive scaling constant introduced to keep the logarithm arguments strictly positive and to improve the linearity of the relationship between X and true depth. Equation (1) was evaluated only over water pixels, identified via a binary land/cloud mask obtained by histogram-based thresholding of the image followed by morphological dilation; the water mask was taken as the logical complement of this land/cloud mask.

Because X is also sensitive to sun-glint and atmospheric scattering at the water surface, a correction step was applied prior to the final depth estimate. Over a homogeneous, presumed-featureless offshore area, a linear regression was fitted between the near-infrared (NIR) band — which is strongly absorbed by water and therefore assumed to carry no seafloor signal, only a surface-reflection component — and each of the blue, green, and red bands. The fitted slope was then used to subtract the surface-reflection contribution from each visible band:

$$B_{\text{corrected}} = B_{\text{original}} - \text{slope} \cdot (\text{NIR} - \min(\text{NIR})) \tag{2}$$

The bathymetric index (Eq. 1) was recomputed using the corrected blue and green bands. Finally, a linear (affine) transformation was applied to convert the corrected index into an approximate depth in metres:

$$Z = a_z \cdot X + b_z \tag{3}$$

with the coefficients a_z, b_z chosen such that the resulting depth range spans $Z_{\min} = 0\,\text{m}$ to $Z_{\max} = 10\,\text{m}$. This calibration is a relative, range-based scaling rather than a fit to independent in-situ depth measurements (e.g. LiDAR bathymetry or sounding data); the resulting depth values should therefore be interpreted as a relative bathymetric index rather than a validated absolute measurement, a limitation discussed further in Section 5.

3.2 Multi-resolution image fusion (Landsat 8 OLI)

The multispectral composite (blue, green, red) was first upsampled by the resolution ratio between the panchromatic and multispectral products, producing a resampled composite, COUL_se, at the same pixel dimensions as the panchromatic band. Two pan-sharpening methods were then applied and compared.

Brovey transform. A synthetic low-resolution panchromatic image, PAN_m, was obtained by averaging the three bands of COUL_se. A linear regression between PAN_m and the true panchromatic band, PAN, was then used to generate a radiometrically consistent high-resolution synthetic panchromatic image, PAN_p. The fused image was computed as:

$$\text{COUL_f}(i,j) = \frac{\text{COUL_se}(i,j)}{\text{PAN_m}(i,j)} \times \text{PAN_p}(i,j) \tag{4}$$

which rescales each colour band proportionally to its contribution to the simulated panchromatic signal, injecting the spatial detail of PAN_p while preserving the relative colour balance of the original composite.

HSV-based fusion . COUL_se was converted to Hue-Saturation-Value (HSV) colour space, separating chromatic information (H, S) from intensity (V). A linear regression between the V channel and the panchromatic band was used, analogously to the Brovey case, to derive a radiometrically consistent high-resolution synthetic V channel, PAN_p, which replaced the original V band. The modified HSV image was then converted back to RGB to obtain the final fused image.

The two methods (Pohl and van Genderen, 1998) were compared qualitatively on the resulting fused composites; no quantitative fusion-quality metric (e.g. ERGAS, spectral angle between fused and reference spectra) was computed, a limitation discussed in Section 5.

3.3 Spectral classification (Sentinel-2 MSI, Camargue)

Three land-cover classes — soil, vegetation, and water — were defined for the Camargue scene. For each class, a region of interest (ROI) was manually selected via `cv2.selectROI()` and saved to disk (NumPy `.npy` format) to allow reuse without repeated manual selection. Each ROI is a multi-band image; the mean reflectance value across all pixels within the ROI was computed independently for each of the four spectral bands, yielding a mean spectral signature per class, $\bar{r}_k$.

Each pixel in the full scene, x, was then assigned to the class whose signature was most similar, under two alternative similarity measures.

Euclidean distance.

$$d(x, r_k) = \sqrt{\sum_{i=1}^{N} \left(x(i) - \bar{r}_k(i)\right)^2} \tag{5}$$

Spectral Angle Mapper (SAM) (Kruse et al., 1993). SAM instead compares the angle between the pixel's spectral vector and each class signature, making it sensitive to spectral shape rather than absolute magnitude:

$$\alpha(x, r_k) = \cos^{-1}\left(\frac{\sum_{i=1}^{N} x(i)\,\bar{r}_k(i)}{\sqrt{\sum_{i=1}^{N} x(i)^2} \cdot \sqrt{\sum_{i=1}^{N} \bar{r}_k(i)^2}}\right) \tag{6}$$

For both measures, each pixel was assigned the class label minimizing d (Eq. 5) or α (Eq. 6), and the resulting label map was rendered as a false-colour classified image by assigning a distinct RGB colour to each class. No quantitative accuracy assessment (e.g. confusion matrix against independently labelled reference pixels) was performed; classification quality was assessed visually by comparing the spatial coherence of the two resulting maps.

3.4 Anomaly detection against a spatially varying background (Sentinel-3 OLCI)

A land/cloud mask was obtained by histogram-based thresholding of the composite image, followed by morphological dilation to close small gaps; the water mask was taken as its logical complement, following the same approach as in Section 3.1.

Within the water mask, the Maximum Chlorophyll Index (MCI) was computed using three OLCI bands centred at 681 nm, 709 nm, and 754 nm (bands 10, 11, and 12), which bracket the chlorophyll-a fluorescence peak:

$$\mathrm{MCI} = \rho'(709) - \left[\rho'(681) + \big(\rho'(754) - \rho'(681)\big) \cdot \frac{709 - 681}{754 - 681}\right] \tag{7}$$

where $\rho'(\lambda)$ denotes top-of-atmosphere or surface reflectance at wavelength λ; Eq. 7 is equivalent to measuring the height of the reflectance peak at 709 nm above the baseline interpolated linearly between the 681 nm and 754 nm bands (Gower et al., 2005).

To distinguish local algae-related anomalies from slowly varying background variation in ocean colour (due to, e.g., sub-surface currents or illumination gradients), a background estimate was computed by applying a large-kernel (51×51 pixel) median filter to the MCI image, restricted to water pixels. The differential index was then obtained as:

$$\mathrm{dMCI} = \mathrm{MCI} - \mathrm{MCI}_{\mathrm{background}} \tag{8}$$

The resulting dMCI map was visualized using a false-colour scale, with the land/cloud mask overlaid to exclude non-aquatic areas from interpretation. The choice of a 51×51 kernel — and the assumption that background ocean-colour variation is adequately captured at that spatial scale while algae patches are not — was not validated against an independent reference and is discussed as a limitation in Section 5.

4 Results

4.1 Bathymetric retrieval (Pléiades)

Figure 1 shows the uncorrected bathymetric index X (Eq. 1) computed over the Porquerolles water mask. The index exhibits a visible radial gradient consistent with increasing depth away from the coastline, but the offshore, presumed-deep area is contaminated by high-frequency noise attributable to sun-glint and atmospheric scattering, and the land/cloud mask appears as a uniform maximum-value region rather than being excluded from the colour scale.

The linear regression between the near-infrared band and each visible band, fitted over a homogeneous offshore area (Figure 2), showed a clear positive linear relationship for all three bands, with the blue band exhibiting the highest reflectance values, followed by green and then red — consistent with the expected order of water penetration depth for these wavelengths. This relationship was used to apply the surface-reflection correction (Eq. 2).

After correction, the marine area appears visibly more homogeneous, particularly offshore, indicating that the parasitic surface-reflection component was substantially reduced (Figure 3). The

bathymetric index recomputed from the corrected bands (Figure 4) shows warm tones (red, orange, yellow) concentrated in shallow coastal areas and cool tones (dark blue) in deeper offshore areas, with a coherent, gradually widening transition around the coastline consistent with a gently sloping coastal shelf.

The final depth map, obtained by applying the affine calibration (Eq. 3) with $Z_{\min} = 0\,\mathrm{m}$ and $Z_{\max} = 10\,\mathrm{m}$, is shown in Figure 5. The spatial pattern — shallow depths near the shoreline, increasing gradually offshore — is physically plausible for the study area, but, as noted in Section 3.1, the absolute depth values have not been validated against independent in-situ measurements and should be interpreted as a relative bathymetric index rather than a calibrated metric quantity.

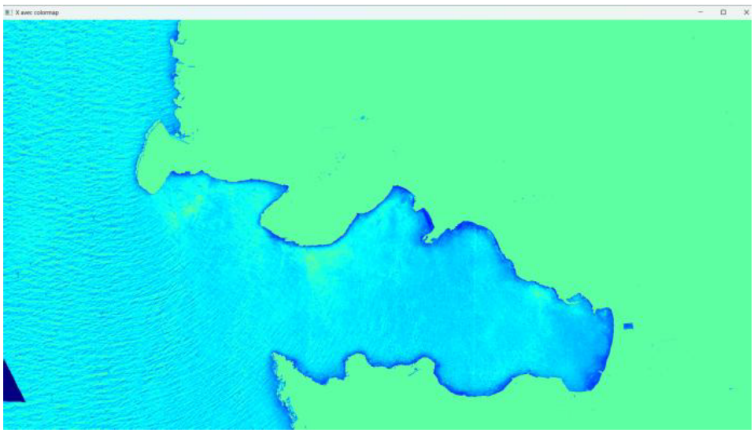


Figure 1: Uncorrected bathymetric index X over the Porquerolles water mask.

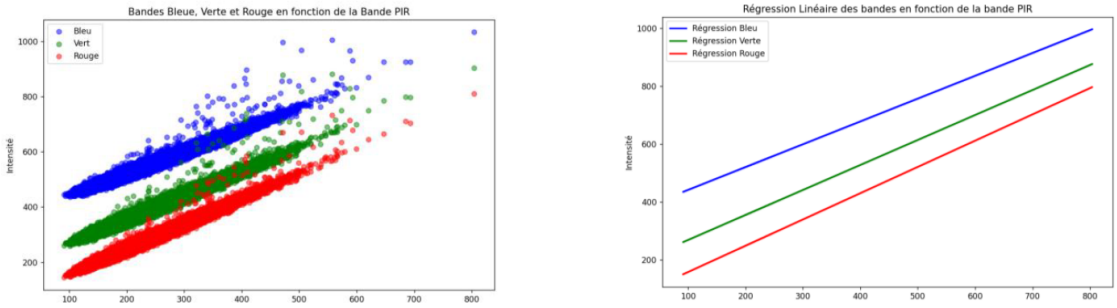


Figure 2: Left: blue, green, and red band reflectance as a function of the NIR band over the offshore reference area. Right: fitted linear regressions used for surface-reflection correction.

4.2 Multi-resolution image fusion (Landsat 8 OLI)

Both fusion methods produced a colour composite at the spatial resolution of the panchromatic band, visibly sharper than the original resampled composite COUL_se (Figure 6). The Brovey fusion result (Figure 7) injects fine spatial detail from the panchromatic band while broadly preserving the tonal contrasts of the original composite. The HSV-based fusion result (Figure 8) likewise transfers spatial detail into the composite, and was judged, on visual inspection, to more closely preserve the original colour balance than the Brovey result, consistent with the HSV method's explicit separation of chromatic (H, S) and intensity (V) information prior to detail injection.

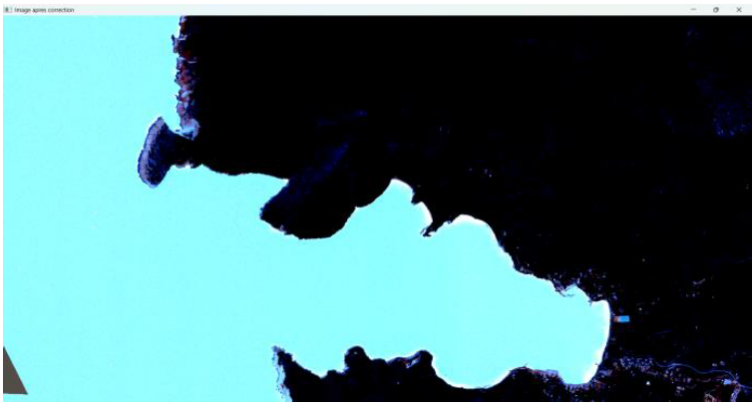


Figure 3: Visible-band composite after surface-reflection correction.

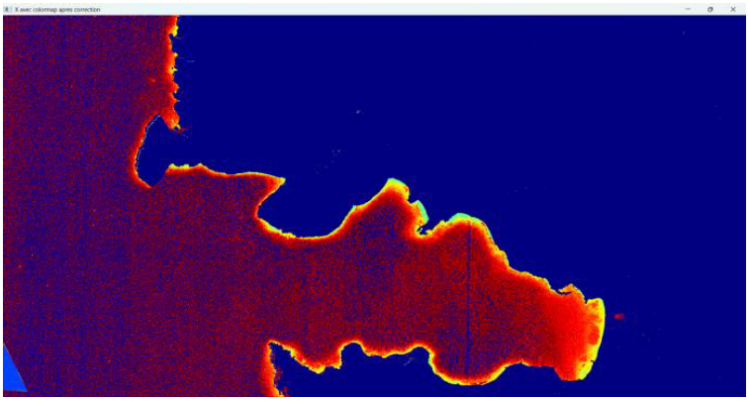


Figure 4: Bathymetric index X recomputed from corrected bands.

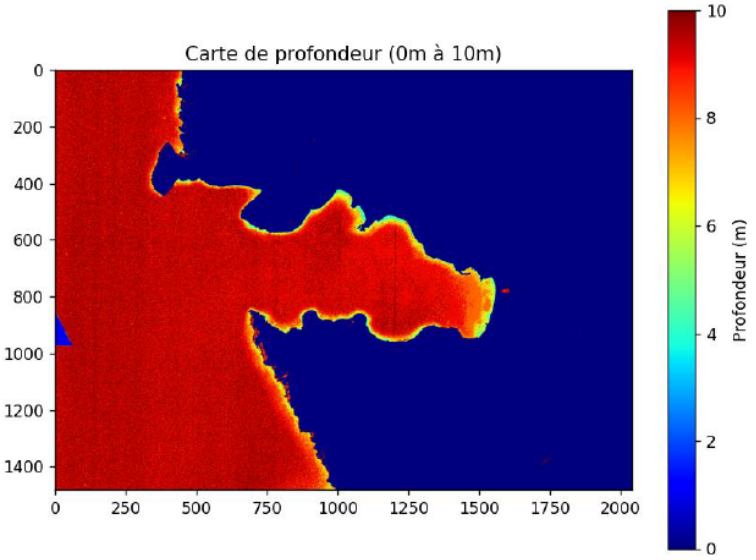


Figure 5: Final estimated depth map (0–10 m), obtained via affine calibration of the corrected bathymetric index.

No quantitative fusion-quality index (e.g. spectral angle or ERGAS between fused and reference imagery) was computed to support this comparison; the assessment above is therefore qualitative only, a limitation discussed in Section 5.

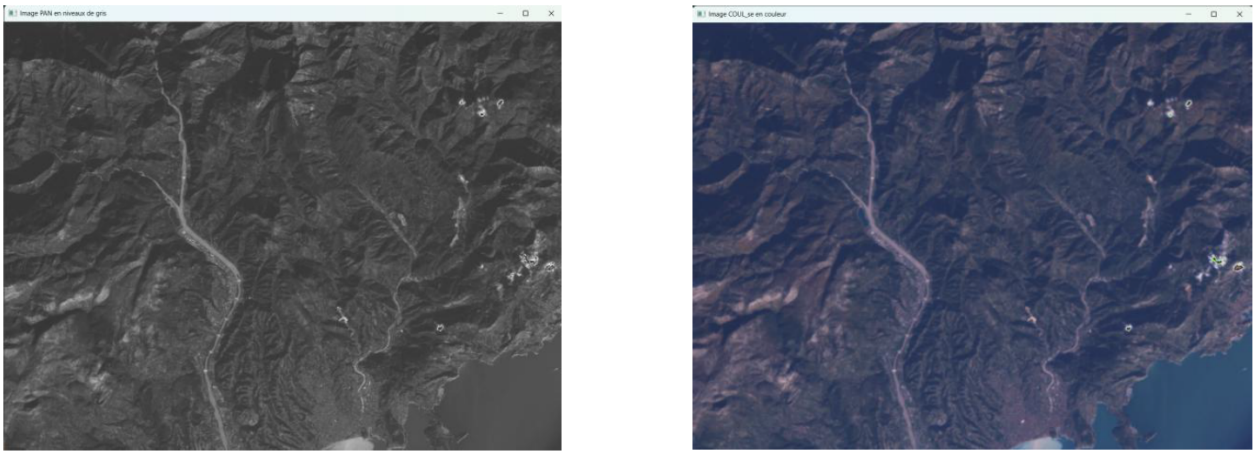


Figure 6: Left: panchromatic band. Right: upsampled multispectral composite COUL_se, prior to fusion.

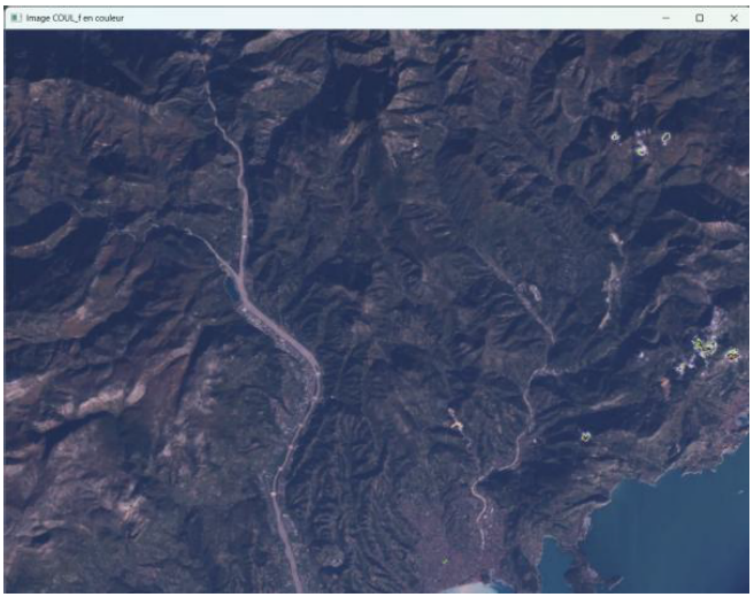


Figure 7: Fused composite obtained via the Brovey transform.

4.3 Spectral classification (Sentinel-2 MSI, Camargue)

The mean spectral signatures extracted from the three ROIs (Figure 9) show distinct behaviour across the four bands. Vegetation exhibits low reflectance in blue and red, consistent with chlorophyll absorption, and a sharp increase in the near-infrared, consistent with the cellular structure of healthy plant tissue. Soil shows a more gradual, monotonic increase in reflectance from blue to near-infrared, without the vegetation red-edge rise. Water shows the highest reflectance in blue,

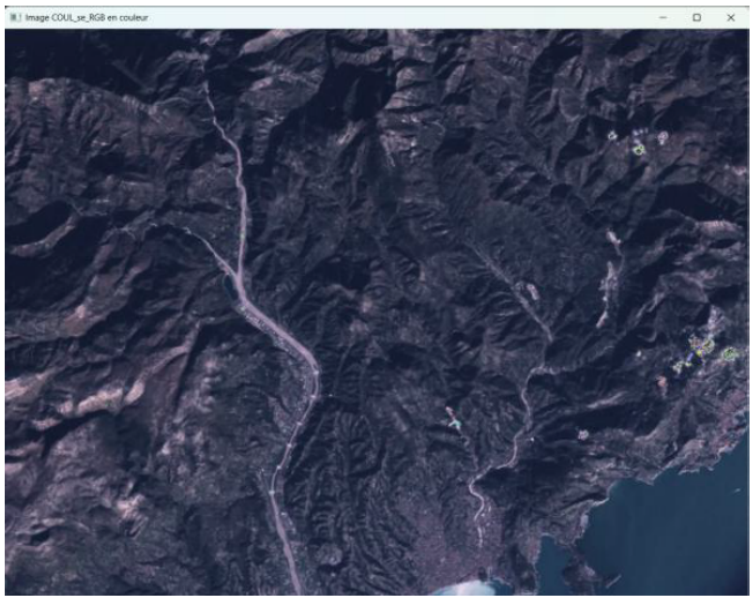


Figure 8: Fused composite obtained via HSV-based substitution.

decreasing progressively toward the red and reaching its lowest value in the near-infrared, consistent with strong water absorption at longer wavelengths.

Classification maps obtained using the Euclidean distance and the Spectral Angle Mapper (Figure 10) show broadly similar large-scale spatial patterns, but differ in local detail. The Euclidean-distance classification produces larger, more homogeneous class regions, but visibly confuses soil and vegetation in transitional zones. The SAM classification is more sensitive to small-scale spectral variation, better delineating water bodies and vegetation, but produces a visibly noisier, more fragmented map. As noted in Section 3.3, this comparison is based on visual inspection only; no confusion matrix or independently labelled validation set was used to quantify classification accuracy for either method.

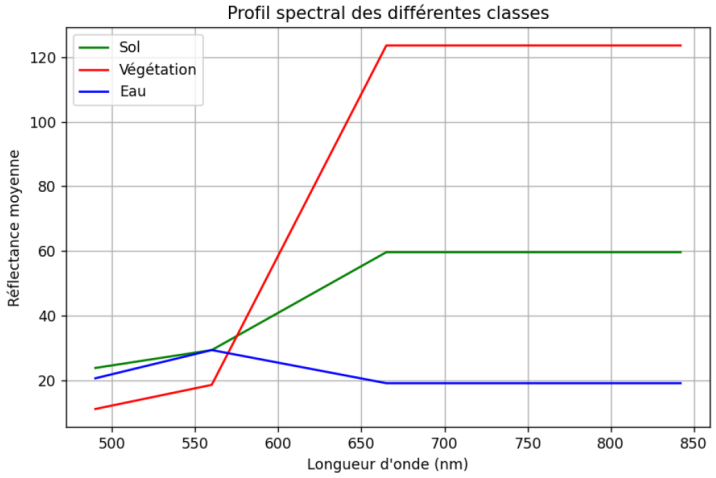


Figure 9: Mean spectral signatures (reflectance vs. wavelength) for the soil, vegetation, and water ROIs.

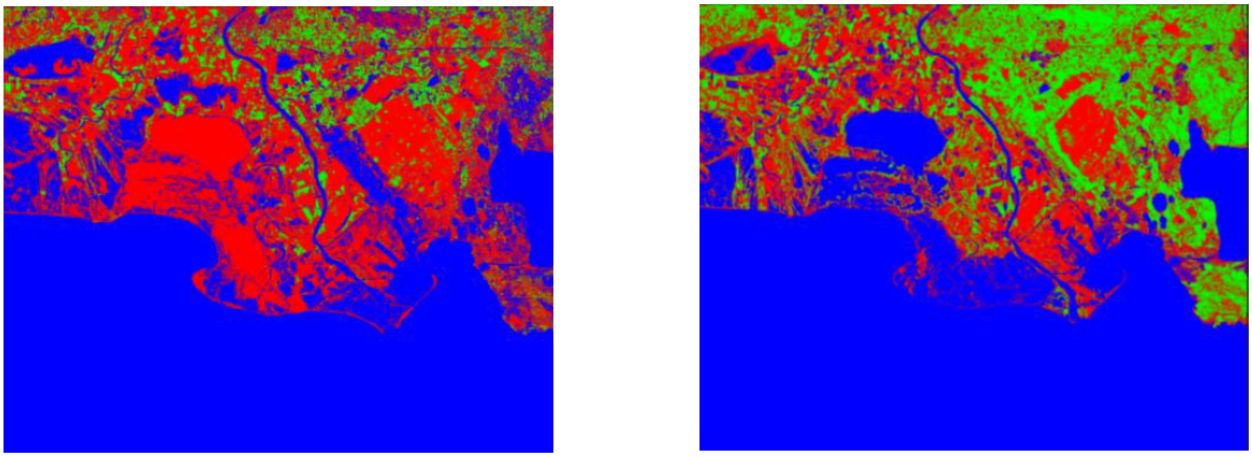


Figure 10: Classification of the Camargue scene using Euclidean distance (left) and the Spectral Angle Mapper (right).

4.4 Anomaly detection against a spatially varying background (Sentinel-3 OLCI)

The MCI computed over the water mask (Figure 11) shows a predominantly low, spatially smooth signal, with a small number of higher-value patches. Because the MCI alone conflates broad, naturally occurring background variation with localized anomalies, a smoothed background estimate was computed (Figure 12) and subtracted from the MCI to obtain the differential index dMCI (Figure 13).

The resulting dMCI map reveals swirling, filamentary patterns of elevated signal, spatially consistent with transport by surface currents, superimposed on a spatially homogeneous background where no anomaly is detected. These patterns are interpreted as probable floating-algae accumulations, on the basis of their spectral association with the chlorophyll-a fluorescence peak captured by the MCI formulation. Land and cloud areas, overlaid in the figure, are excluded from this interpretation by construction, since the index is computed and displayed only within the water mask. As with the other case studies, no independent (e.g. in-situ or higher-resolution) reference was available to confirm that the detected anomalies correspond to actual algae presence rather than another source of localized reflectance variation.

5 Discussion

5.1 Synthesis across case studies

The four case studies share a common underlying logic despite differing in sensor, target variable, and output type: a physically or spectrally motivated transformation (a band ratio, a fusion rule, a distance or angle metric, a background subtraction) is applied to raw multi-band imagery to recover a signal that is not directly observable in the raw bands themselves — respectively, depth, spatial detail, thematic class, and a localized biological anomaly. In each case, a masking step (land/cloud exclusion, water isolation) was used to restrict the computation to the pixels for which the underlying physical assumptions hold, which is a methodologically appropriate step common to all four workflows.

The bathymetry and algae-detection case studies further share a specific technique: separating a signal of interest from a spatially varying nuisance component using a reference derived from the same or an auxiliary band (the NIR-based surface-reflection correction in Section 3.1; the median-filtered background in Section 3.4). This recurring "estimate and subtract the background" strategy is a

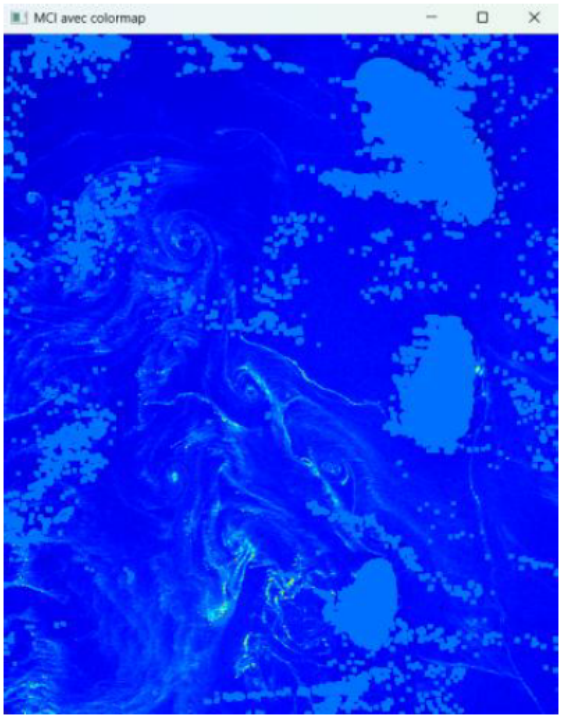


Figure 11: MCI computed over the water mask.

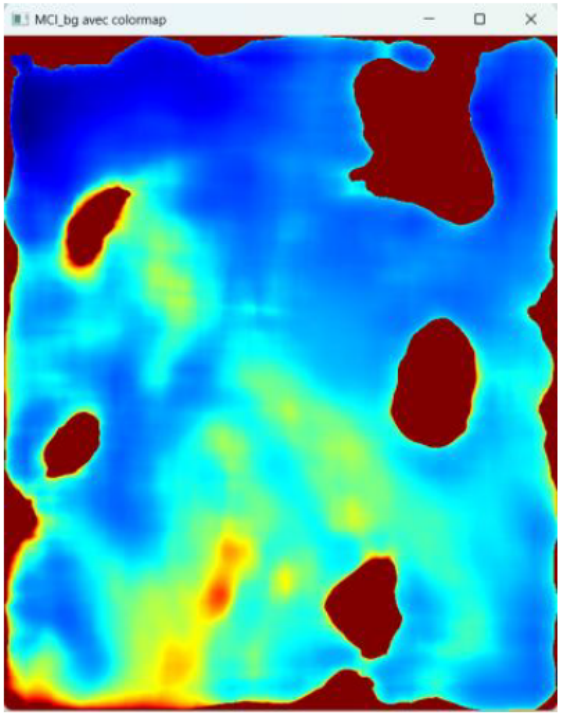


Figure 12: Smoothed MCI background, obtained via a 51×51 median filter restricted to water pixels.

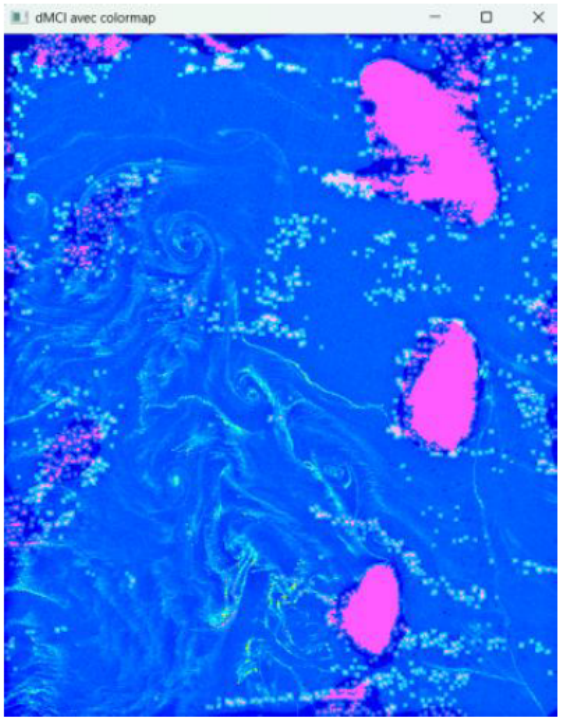


Figure 13: Differential chlorophyll index (dMCI), with land/cloud mask overlaid, highlighting local anomalies attributed to floating algae.

genuinely transferable analytical pattern, applicable in principle to other settings where a target signal must be separated from slowly varying or spatially correlated background variation.

5.2 Strengths

The case studies demonstrate a coherent, reproducible processing pipeline implemented directly in Python rather than relying solely on graphical remote-sensing software, which supports transparency and reusability of the workflow. The bathymetry and fusion case studies in particular go beyond simple thresholding: both involve a regression-based correction step calibrated on the data itself (the NIR–visible band regression in Section 3.1; the PAN–composite regression in Section 3.2), which is a more principled approach than applying a fixed, un-calibrated correction. The classification case study's use of two conceptually distinct similarity measures (Euclidean distance and SAM) and explicit comparison of their respective behaviour (homogeneous but confusable classes vs. detail-sensitive but noisier classes) shows an appropriate awareness that no single classifier is universally preferable, and that the choice depends on the class structure and objectives of the task at hand.

5.3 Limitations

Absence of independent validation. None of the four case studies were validated against an independent reference dataset. The bathymetric depth map (Section 4.1) was not compared against in-situ soundings, LiDAR bathymetry, or a nautical chart; the land-cover classification (Section 4.3) was not assessed via a confusion matrix against independently labelled pixels; the fusion outputs (Section 4.2) were not evaluated using a standard fusion-quality metric (e.g. ERGAS, spectral angle between fused and reference spectra); and the floating-algae anomalies (Section 4.4) were not cross-

checked against a higher-resolution image or field observation. In each case, the reported results should therefore be read as methodologically plausible outputs of the applied technique, not as independently confirmed measurements.

Unvalidated calibration and parameter choices. The final depth calibration (Eq. 3) fixes $Z_{\min} = 0\,\text{m}$ and $Z_{\max} = 10\,\text{m}$ by assumption rather than by regression against known depth values, meaning the resulting map is best interpreted as a relative bathymetric index rescaled to a plausible depth range, rather than a calibrated absolute measurement. Similarly, the 51×51 median-filter kernel used to estimate the MCI background (Section 3.4) was not selected via a sensitivity analysis or comparison against alternative kernel sizes, and the original material notes analogous, unquantified sensitivity of the flood and burn-scar thresholds (not detailed in this paper) to the chosen threshold value.

Manual and subjective steps. Threshold selection (for the land/cloud masks in Sections 3.1 and 3.4) and ROI selection (for the classification signatures in Section 3.3) were performed manually based on visual inspection of histograms and imagery, rather than via an automated, reproducible criterion (e.g. Otsu's method for thresholding, or a stratified random sampling scheme for ROI placement). This introduces a degree of subjectivity and limits the exact reproducibility of the reported results by a third party working from the same raw imagery.

5.4 Directions for more rigorous follow-up work

The limitations above point to concrete, achievable improvements rather than requiring a change of approach. Depth retrieval (Section 3.1) could be calibrated against even a small set of independent depth measurements (e.g. chart soundings at a handful of known points), converting the current relative index into a validated metric estimate. Threshold selection could be automated using a standard statistical criterion (e.g. Otsu's method) and its sensitivity quantified by reporting how the derived output (flooded area, burned area, depth range) varies across a range of threshold values, rather than selecting a single value by inspection. Classification accuracy (Section 3.3) could be assessed quantitatively via a confusion matrix computed against a small independently labelled validation set, and fusion quality (Section 3.2) via a standard index such as ERGAS. Finally, the background-subtraction approach used for algae detection (Section 3.4) could be extended by testing sensitivity to the median-filter kernel size, and, where available, cross-referenced against a higher-resolution optical image acquired on a similar date.

5.5 Toward a georeferenced, vector-aware extension of this pipeline

The four case studies presented above operate entirely on raster arrays in image (row, column) coordinates, using NumPy and OpenCV; no explicit coordinate reference system (CRS), reprojection, or vector data layer is handled at any point in the pipeline. This is a genuine limitation, in particular for the bathymetric case study (Section 3.1), where a calibrated depth product is only usable downstream once every pixel has a real-world coordinate and can be related to independent reference data (e.g. nautical chart soundings, or a georeferenced shoreline vector for masking).

A natural extension, not implemented here but sketched below at the level of the concrete libraries and operations involved, would replace the current NumPy/OpenCV raster handling with `rasterio` (built on GDAL) for CRS-aware I/O, and introduce `geopandas` for vector operations. Practically, this would involve:

1. Reading each band with `rasterio.open()` rather than `cv2.imread()`, preserving the affine transform and CRS (e.g. UTM 31N for the Porquerolles scene) attached to the source product, so that every pixel index can be converted to a real-world coordinate via `rasterio.transform.xy()`.

2. Reprojecting or resampling bands to a common grid using `rasterio.warp.reproject()` when combining products of differing native resolution or CRS (relevant to the fusion case study, Section 3.2, where the panchromatic and multispectral bands would be aligned on an explicit georeferenced grid rather than by array upsampling alone).

3. Converting the binary land/cloud and water masks (Sections 3.1 and 3.4) from raster masks to vector polygons using `rasterio.features.shapes()`, then loading the result into a `geopandas.GeoDataFrame` to enable standard vector operations — computing shoreline length, polygon area in a projected CRS rather than pixel counts, or spatial joins against independent reference layers (e.g. a charted coastline or bathymetric survey extent) for validation.

4. For the classification case study (Section 3.3), exporting the classified raster to vector polygons per class in the same way, enabling direct area computation in hectares from polygon geometry (in a projected CRS) rather than from a pixel count times a nominal ground sample distance, and making the result directly usable as a GIS layer.

This extension would not change the underlying spectral or physical methods described in Section 3; it would instead embed them within a standard, reusable geospatial data model, and provide the concrete missing link needed to validate the current results against external reference data (Section 5). This gap, and the concrete route to closing it, was identified rather than resolved here for reasons of data availability rather than methodological difficulty.

6 Conclusion

This paper presented four case studies in quantitative and categorical information extraction from optical satellite imagery: depth retrieval from Pléiades multispectral imagery via a corrected band-ratio index; spatial-detail injection into a Landsat 8 OLI multispectral composite via Brovey and HSV-based pan-sharpening; land-cover classification of a Sentinel-2 MSI scene using Euclidean-distance and Spectral Angle Mapper similarity measures; and detection of localized chlorophyll-a anomalies in Sentinel-3 OLCI imagery via background subtraction. Despite differing in sensor, spatial resolution, and target variable, all four workflows follow a common structure: masking the imagery to the domain over which the underlying physical or spectral assumptions hold, constructing an index or reference signature appropriate to the target variable, and, where relevant, correcting for a nuisance signal component using a regression- or filtering-based estimate derived from the data itself.

The results show that meaningful depth, resolution, classification, and anomaly information can be recovered from single-date optical imagery using comparatively simple, computationally inexpensive techniques implemented in a standard Python geospatial stack. At the same time, the case studies share a set of limitations that should temper how their outputs are read: none were validated against an independent reference dataset, several parameter choices (depth-range calibration, filter kernel size, binarization thresholds, ROI placement) were made by visual inspection rather than by an automated or externally justified criterion. These are not fundamental flaws in the chosen methods but concrete, addressable gaps — automated thresholding, quantitative accuracy and fusion-quality assessment, and calibration against independent reference measurements — that would strengthen the reliability of the derived products if this work were extended.

Beyond the four case studies detailed here, the broader practical training also covered bi-temporal flood-extent mapping, burn-scar mapping, vessel counting via connected-component analysis, and field-spectrum acquisition and classification (Section 1), reinforcing the same core competencies — spectral index construction, image thresholding, bi-temporal comparison, and per-pixel classification — across a wider range of sensors and application contexts. Taken together, this body of work illustrates a practical, hands-on grounding in deriving quantitative and categorical geospatial information from heterogeneous, resolution-limited optical imagery: extracting a physical measurement (depth) from an indirect spectral proxy, combining data of differing spatial resolution into a single higher-resolution product, and separating a signal of interest from natural background variation. These are precisely the kinds of methodological building blocks required when measuring a physical or structural quantity from imperfect, multi-source, and often data-scarce remote-sensing observations — a problem setting directly relevant to further doctoral-level research on quantitative urban or environmental measurement from Earth-observation data.

Reproducibility

Code and all raw result CSVs, and figures are available at: https://github.com/AboOraig/sentinel-multispectral-analysis.

References

Gower, J., King, S., Borstad, G., and Brown, L. (2005). Detection of intense plankton blooms using the 709 nm band of the MERIS imaging spectrometer. *International Journal of Remote Sensing*, 26(9):2005–2012.

Kruse, F. A., Lefkoff, A. B., Boardman, J. W., Heidebrecht, K. B., Shapiro, A. T., Barloon, J. P., and Goetz, A. F. H. (1993). The spectral image processing system (SIPS)—interactive visualization and analysis of imaging spectrometer data. *Remote Sensing of Environment*, 44(2–3):145–163.

Lyzenga, D. R. (1978). Passive remote sensing techniques for mapping water depth and bottom features. *Applied Optics*, 17(3):379–383.

Pohl, C. and van Genderen, J. L. (1998). Multisensor image fusion in remote sensing: concepts, methods and applications. *International Journal of Remote Sensing*, 19(5):823–854.

Stumpf, R. P., Holderied, K., and Sinclair, M. (2003). Determination of water depth with high-resolution satellite imagery over variable bottom types. *Limnology and Oceanography*, 48(1, part 2):547–556.